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THERMAL PROPERTY MEASUREMENT OF THIN FIBERS – A DIRECT APPROACH

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ABSTRACT

Fiber thermal characterization is often accomplished by indirect means, such as embedding the fiber in a matrix, measuring the thermal response of the composite, and relating for the contributions of the fiber and matrix to the overall behavior or measuring bundles of fibers. To improve the accuracy of the composite-based or bundle-based techniques, several different contact (hot wire and dc thermal bridge) and non-contact (Raman shift and IR thermography) methods have been developed to directly measure the thermal properties of individual fibers. To improve on the shortcomings of these methods, this paper presents the experimental results of an improved transient electrothermal (TET) method, as well as a 3ω-based method that better accounts for all sources of heat transfer, particularly heat loss by radiation. The incorporation of radial radiation heat loss becomes a significant factor as the size of the fibers decrease. This work describes practical applications of the methods to measure the properties of the fibers, including sample preparation for electrically conductive and non-conductive samples, data acquisition and calibration, data analysis, and sample property determination.

Results include validation of the methods with electrically conductive (platinum) and non-conductive (glass) fibers to improve upon the initial validation of the generalized electrothermal method which focused only on short, conductive fibers. The axial thermal conductivity and diffusivity of several high performance fibers are presented. The novelty of this paper is that it serves as both a compilation of previous research on the transient electrothermal and 3ω methods [1–6], measurements of new silk fibers, and practical information associated with the methods that improve the accuracy of the measured thermal property, as well as presenting thermal properties of additional fibers (carbon fiber and natural and synthetic spider silks).

To improve upon the long sample preparation time required for the TET and 3ω methods, future work focused on the development of a quantum dot-based photothermal fluorescence method is presented.

INTRODUCTION

Because thin fibers are used in many materials, most noticeably composites, the characterization of individual fibers for both thermal and mechanical properties is important. To measure the thermal conductivity and diffusivity of thin fibers, several methods have been developed including direct methods (defined by contact with the fiber for measurement, such as the dc thermal bridge [7], hotwire method [8], 3ω [9], and transient electrothermal or TET [10]) and indirect methods (defined by non-contact sensing of the fibers IR lock-in thermography [11] and Raman shift [12]). Several difficulties arise when attempting to measure the thermal behavior of nano- and micro-sized fibers because of small dimensions, accurate temperature measurements, and fully accounting for all the heat transfer mechanisms that occur, while trying to make all of these behaviors conform to an expected thermal model. These difficulties can be shown where the thermal conductivities of the dragline of two spiders of different species have a 400x discrepancy based on two different methods (IR thermography and TET [13,14]), and significant work has been done on the 3ω and TET methods to match theory with experiment to

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improve their accuracy [1–6] to account for this discrepancy. This paper presents an overview of the relevant work on these methods and describes practical applications to measure the properties of the fibers, including sample preparation for electrically conductive and non-conductive samples, data acquisition and calibration, data analysis, and sample property determination.

THERMAL MODELS

A summary of the relevant models is presented (while derivation for the presented models is detailed in [2] for 3ω and in [3] for TET) with the final models that relate system responses to properties. The basis for these models is a 1D transient solution of the heat equation solved for the average axial temperature of the fiber due to Joule heating and with constant temperature heat sinks. The samples are placed in a vacuum chamber, exposed to an electrical source, and the response is measured by an electrical meter, as shown schematically in Fig. 1.

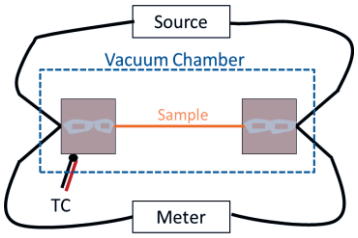


Figure 1. Simplified schematic of measurement operation.

TET Method

For the TET method, the electrically conductive portion of the fiber (either the fiber itself or a deposited coating) is exposed to a constant current source step input and the average voltage response is then measured over time, where the transient response is related to the thermal diffusivity and the steady state response is related to the thermal conductivity. These voltage responses are linearly related to the resistance response of the sample, which is in turn related to the temperature response of the fiber. A resistance-temperature calibration must be performed to relate the measured resistance to the average temperature experienced, and must be performed on each individual fiber. Expressions are developed for thermal conductivity and thermal diffusivity, each with two forms - a full model and a reduced model. The full models include an effective heat transfer coefficient (H_e^2 – Eq. 1, where $h_r=4\varepsilon\sigma T_0^3$) that considers the radial convection and radiation heat transfer (similar to a fin) and non-constant heating (changes in resistance due to temperature rise that then affect the amount of Joule heating experienced by the fiber). As the diameter of the fiber decreases or the length increases, these effects become more significant. The full models are Eqs. 2 and 3, while the reduced models are Eqs. 4 and 5, for thermal conductivity and thermal diffusivity, respectively. The reduced models match the full models when $H_e=0$. Additionally, a modified reduced model for both conductivity and diffusivity is given as Eqs. 6 & 7 ($e_{kb}=0.1L^2H_e^2$), where this modified model is based on a numerical experiment in [3], which related the error of using the reduced model compared to the full model. The error expression is shown graphically in Fig. 2 and allows the researcher to use the reduced model with a correction factor when the emissivity of the sample is known (or determined). Properties are determined directly by the equations for conductivity or curve fit based on a non-linear Levenberg-Marquardt algorithm for diffusivity.

$$H_e^{\,2} = \frac{(h_r + h_c)}{Dk} - \frac{4I^2 R_0 \alpha_T}{\pi D^2 L k} \tag{1}$$

$$k = \frac{4I^2 R_0 R' L}{D^2 \pi (R_S - R_0)} \frac{2 - 2\cosh(LH_e) + LH_e \sinh(LH_e)}{(LH_e)^3 \sinh(LH_e)} \tag{2}$$

$$\frac{\Delta T(t)}{\Delta T_s} = 1 - \frac{8(LH_e)^3 \sinh(LH_e)}{2 - 2\cosh(LH_e) + LH_e \sinh(LH_e)} \times \sum_{n=1}^{\infty} \frac{e^{-\left[(2m-1)^2\pi^2 + (LH_e)^2\right]\alpha t/L^2}}{\left[(2m-1)\pi\right]^4 + \left[LH_e(2m-1)\pi\right]^2} \tag{3}$$

$$k = \frac{I^2 R_0 R' L}{3\pi D^2 (R_s - R_0)} \tag{4}$$

$$\frac{\overline{\Delta T}}{\overline{\Delta T_s}} = \frac{\Delta V}{\Delta V_s} = 1 - \frac{96}{\pi^4} \sum_{m=1}^{\infty} \frac{e^{-(2m-1)^2 \pi^2 \alpha t/L^2}}{(2m-1)^4} \tag{5}$$

$$k = \frac{I^2 R_0 R' L}{[3\pi D^2 (R_s - R_0)](1 + e_{kb})} \tag{6}$$

$$\frac{\overline{\Delta T}}{\overline{\Delta T_s}} = \frac{\Delta V}{\Delta V_s} = \left[1 - \frac{96}{\pi^4} \sum_{m=1}^{\infty} \frac{e^{-(2m-1)^2 \pi^2 \alpha t/L^2}}{(2m-1)^4}\right]\left(\frac{1}{1 + e_{kb}}\right) \tag{7}$$

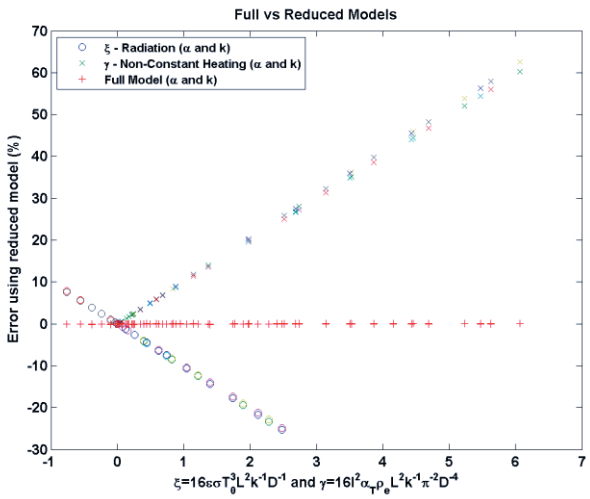


Figure 2. Error of property measurement by using reduced model compared to full model.

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3ω method

For the 3ω method, two types of electrical sources for the initiating of Joule heating are considered, a current source and a voltage source. The derivation of the 3ω method is based on a sinusoidally modulated, dc offset input current at 1ω and then measuring the system's voltage response at the 3ω harmonic. The benefit of the 3ω method is improved signal-to-noise ratio, as well as the ability to simultaneously measure the thermal conductivity, thermal diffusivity, and volumetric heat capacity, based on the probing frequency. Similar to the TET method, a resistance temperature calibration is needed since the voltage response of the fiber is due to the temperature and resistance voltage change at 2ω. The general expression of the 3ω voltage is given in Eq. 8, where $m^2 = i2\omega/\alpha + 4h_r/(Dk)$. This equation can be simplified at the high and low frequency limits and provides expressions for the thermal conductivity and volumetric heat capacity, respectively. These expressions are given in Eqs. 9 and 10.

$$V_3 = \frac{{I_0}^2 R_0 R'[2 - 2\cosh(mL) + mL\sinh(mL)]}{L^2 m^3 k\pi D^2 \sinh(mL)} \quad (8)$$

$$V_3 = \frac{{I_0}^3 R_0 R' L}{12k\pi D^2} \quad (9)$$

$$V_3 = \frac{{I_0}^3 R_0 R'}{i2\varpi\rho C_p \pi D^2 L} \quad (10)$$

	Description	Thermal Diffusivity (10^{-7} m^2/s)								Lit Value	Thermal Conductivity (W/mK)								Lit Value
Synthetic Silk	As Spun	1.68																	
	80:20 MaSp1:MaSp2, 3x water dipped	2.23																	
	80:20 MaSp1:MaSp2, 2x water stretched	2.31																	
Glass Fiber	L^2He^2	*0.04*	*1.18*	*4.81*	*11.21*	*0.26*	*1.75*	*3.31*	*4.39*		*0.04*	*1.18*	*4.81*	*11.21*	*0.26*	*1.75*	*3.31*	*4.39*	
	Reduced	7.43	8.10	9.48	9.51	10.4	10.7	13.4	18.9	8.00	1.35	1.49	2.04	2.78	1.38	1.58	1.81	1.84	1.38
	Full	7.90								8.00	1.35	1.33	1.4	1.29	1.35	1.34	1.37	1.25	1.38
	Modified	7.32	8.16	8.00	7.84	7.69	7.37	8.89	8.21	8.00	1.35	1.33	1.38	1.32	1.35	1.34	1.36	1.28	1.38
	Curve Fit - Reduced	7.60									1.35								
	Curve Fit - Full										1.35								1.38
	Curve Fit - Modified										1.35								
Natural Spider Major Silk	Mean	6.20									1.23								
	Reduced	45.8	12.5	62.4	173	6.50	12.7	1274	415		9.88	2.85	10.96	31.8	1.2	2.32	190.6	66.92	
	Full	5.80	5.50	5.90	5.90	6.10	6.40	6.80	6.60		1.36	1.29	1.14	1.22	1.15	1.19	1.2	1.23	
	Curve Fit - Reduced	6.30									1.18								
Spider Minor Silk (20m/min)	Curve Fit - Reduced	4.00									1.18								
Carbon Fiber (AS4)	Curve Fit - Reduced	62.0																	
	Modified	62.5																	
Platinum	3ω - Current w/o cancelling	255		265		256		255		251	74		75.3		71.8		74.1		72
	3ω - Current w/ cancelling	243		261		250		258		251	71.9		72.9		74.9		74.0		72
	3ω - Voltage w/ cancelling	248		257		255		257		251	120.5		112.8		94.4		94.0		72
	3ω - Voltage w/ cancelling and correction										71.4		73.0		74.5		74.5		72
	TET - Reduced Curve Fit	252								251									
	TET - Full Model										75.3								71.6
		Volumetric Heat Capacity (MJ/m^3K)																	Lit Value
	3ω - Current w/o cancelling																		
	3ω - Current w/ cancelling	2.95				2.82				2.96					2.84				2.85
	3ω - Voltage w/ cancelling																		
	3ω - Voltage w/ cancelling and correction																		
	TET - Reduced Curve Fit																		
	TET - Full Model																		

Table 1. Summary of selected fiber thermal properties by both methods.

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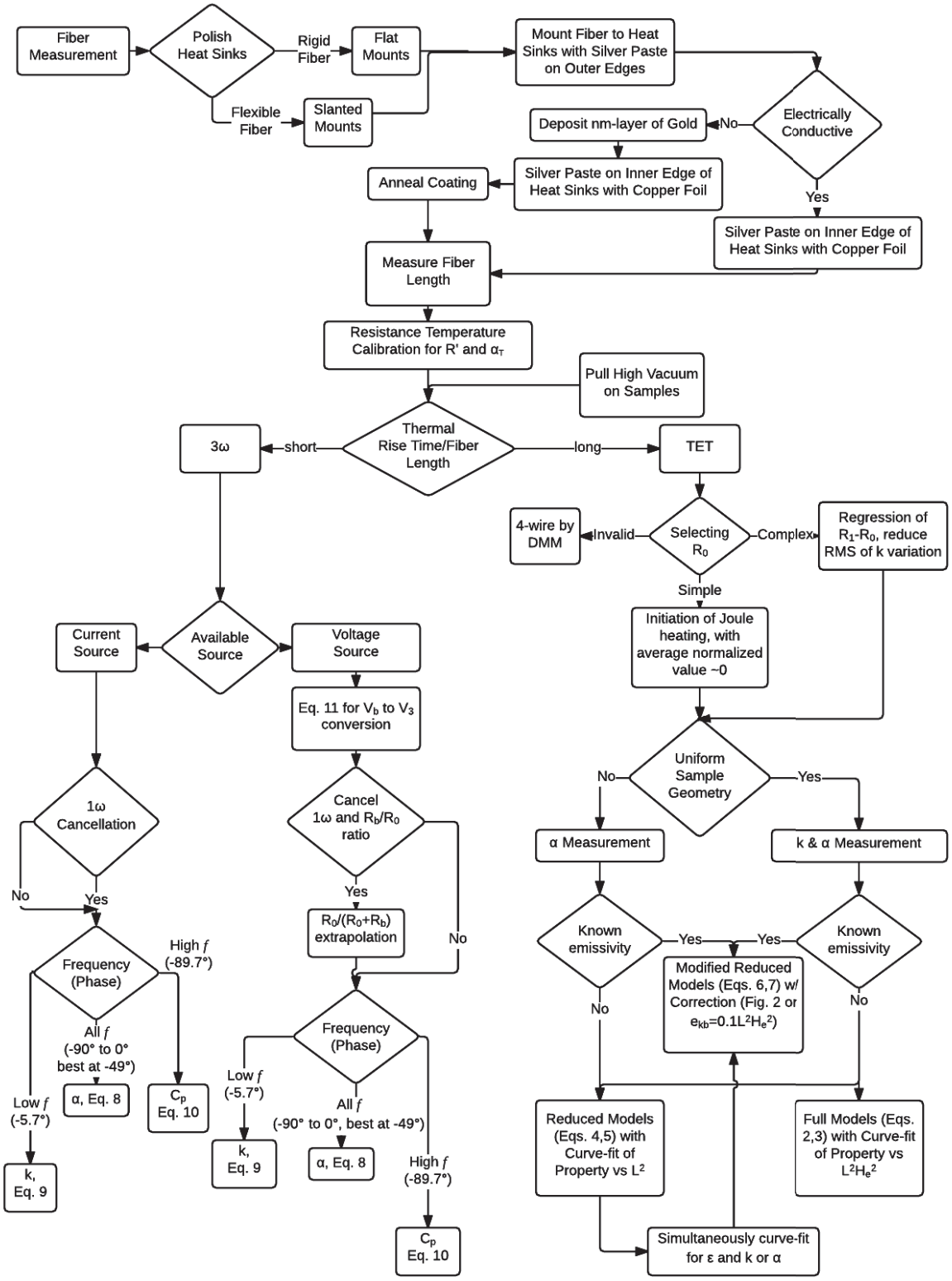


Figure 3. Flow chart for application of 3ω and TET methods for determining thermal properties of fibers.

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All of these models for the 3ω method are based on a constant current source, but a voltage source using a bridge can also be used, with the only modification being that the measured bridge voltage must be related to the constant current voltage by Eq. 11. The 3ω method can be used with or without cancelling of the fundamental harmonic, with more accurate measurements given by the cancellation setups. In the case of a voltage source, this is accomplished by balancing of the whetstone bridge, but presents further complications in the measuring of the thermal properties. This is because the ratio between the resistance of the sample and the resistance of the bridge affects the measured property. In the case of high resistance samples (often encountered with coated samples), this effect is usually negligible, but for small resistance fibers (such as the platinum wires often used to validate models), this ratio must be considered and is used to scale the measured bridge voltage. This scaling/correction factor is given as a linear regression at different resistance ratios – $R_0/(R_b+R_0)$ - and reduces the measured thermal conductivity of platinum from 120 $Wm^{-1}K^{-1}$ to near the literature value of 71 $Wm^{-1}K^{-1}$. Further details are given in [1].

$$V_3 = \frac{R_0 + R_b}{R_b} V_b \tag{11}$$

The next section of this paper presents practical considerations in the application of these different models to a physical system for measuring the thermal properties of thin fibers.

RESULTS

A summary of the results [1–6] of the thermal property measurement of multiple different types of fibers are given in Table 1. The platinum and glass fiber samples provide validation of the models used as the thermal properties measured agree well with bulk values of these materials. The validation of the glass fiber is of significant importance because the majority of thin fiber thermal property measurements that use electric heating only validate the models used with an electrically conductive material like platinum, but do not validate the more complicated behavior that occurs when the sample must be coated.

Additionally, the thermal diffusivity of a Hexcel AS4 carbon fiber, a synthetic spider silk with improved mechanical properties compared to the as spun synthetic silk reported in [6], and the minor ampullate silk of the *N. clavipes* extracted at a rate of 20 m/min is presented for the first time in Table 1. The synthetic spider silk is composed of the same proteins (MaSp1/MaSp2), in the same proportion that exists in the natural dragline spider silk, yet it has a lower thermal diffusivity. This reduction in properties compared to the natural fiber has also been observed in the mechanical properties [15], as is the improvement of both types of properties compared to the as spun fiber because of additional processing of the silk during their production. The minor ampullate natural spider silk has both a lower strength and elasticity than the dragline silk (due to the absence of elastic β-turn spirals structures in the constituent proteins [16]). Again, this decrease in mechanical properties is shown to also happen with the thermal diffusivity.

The uncertainty of the 3ω method for thermal conductivity, thermal diffusivity, and volumetric heat capacity are ~9%, ~4%, and ~11%, respectively, and the uncertainty associated with the TET method are ~7% and ~4% for thermal conductivity and thermal diffusivity, respectively.

PRACTICAL EXPERIMENTAL CONSIDERATIONS FOR ACCURATE PROPERTY MEASUREMENT

Sample Preparation

To aid in the description of practical considerations for fiber measurements by these direct electrical methods, the flow chart in Fig. 3 is given as a guide and will be further described. The process begins by creating the heat sinks, where they are polished flat on three sides by a disc polisher to ensure correct location of fiber terminations, good contact for the fiber onto the heat sink, and a good wetting surface for the wires to be soldered onto. For flexible samples, the surface in contact with the fiber is polished to be slightly slanted so that the higher surfaces are facing each other. Between six to eight (plus a large common) heat sinks are placed on the composite side of a PCB board and are epoxied to the board in a staggered manner so that increasingly longer fibers are suspended between the heat sinks, with the range being between ~1mm and ~15mm, as shown in Fig. 4. Each heat sink has two wires soldered onto the end, and the common heat sink has an additional thermocouple soldered to the other end to ensure that the constant temperature assumption is not invalidated. The fibers are then mounted to the heat sinks with silver paste near the outer edges of the heat sinks. For an electrically conductive fiber, a second glob of silver paste is applied at the same time to the inner edges of the heat sinks and squished down with a piece of copper sheet to ensure that the fiber cleanly leaves the silver paste near the edge of the heat sink. For the non-conductive fibers, this second silver pasting occurs after the coating of 10-15nm of gold onto the samples. Prior to the coating, aluminum foil is placed between the heat sinks and underneath the fibers to ensure electrical contact only through the gold coating on the fiber.

The coating process is then followed by multiple days of annealing to reduce variation of the gold coating's resistance. The coating machine for the current project has an effective area of the gold coating ~1.5in and is the limiting factor on the number of fibers that can be placed on a sample mount. The coating annealing can take almost two weeks when the annealing temperature cannot go above 100 °C, but can be reduced with a higher annealing temperature. Annealing is vitally importance so that the resistance of the fibers does not decrease during the thermal property tests. The effect of this annealing is shown in Fig. 5, where silk samples and glass samples were annealed for several days and at different annealing temperatures. The result is that without annealing, the

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measured thermal property can be several times higher than with annealing, because the annealing causes the resistance of the coating to decrease and the R' value to change. This annealing can unfavorably occur during the testing of the fibers when current is supplied, and results in a lower steady state resistance, which means a lower temperature rise, and (based on Eq. 4), a higher measured thermal conductivity.

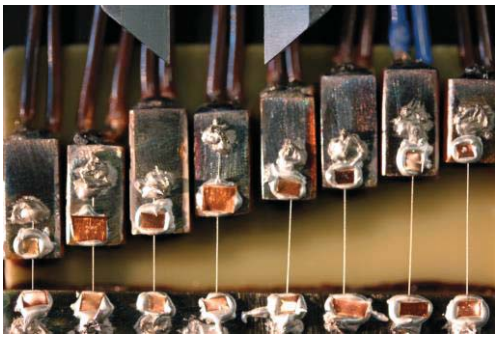


Figure 4. Sample mount image showing differing lengths, pasting locations, caliper for accurate length measurements, and gold coating.

Resistance-Temperature Calibration

The lengths of all fibers on the mount are then measured by means of a camera with a 5mm reference caliper in the picture, and then determined by digital imaging. Difficulty can arise if the silver paste was not applied well, because the location of where the fiber leaves the paste may not be clear [4]. Since the length measurement is a significant source of uncertainty, it is important to have a well-defined exit of the fiber, which is one of the reasons for the copper foil applied during the mounting step. All fibers are then calibrated to get their resistance behavior as a function of temperature, since this is needed to determine the thermal conductivity of the fiber. Examples of good and bad calibrations are given in Fig. 6, with the bad calibrations being indicative of all calibrations done with carbon fibers. The good calibrations are repeatable and show a highly linear trend for the gold coating on a spider silk substrate. Other fibers coated in the same way (after annealing) also show a similar good linear behavior, only the carbon fibers have provided such abnormal behavior, and the reason for this is unknown.

After the calibration is performed and the samples are placed in a high vacuum chamber (<0.001 Pa, to eliminate convection heat losses), the testing and analysis can then travel along two different lines, with the main decision being determined by the time of the thermal response of the fiber. For long fibers or low diffusivity fibers where the TET would measure at most three minutes of rise time, the TET is a better method. For shorter fibers or shorter rise times (less than 1 second), the 3ω method is better suited.

3ω Considerations

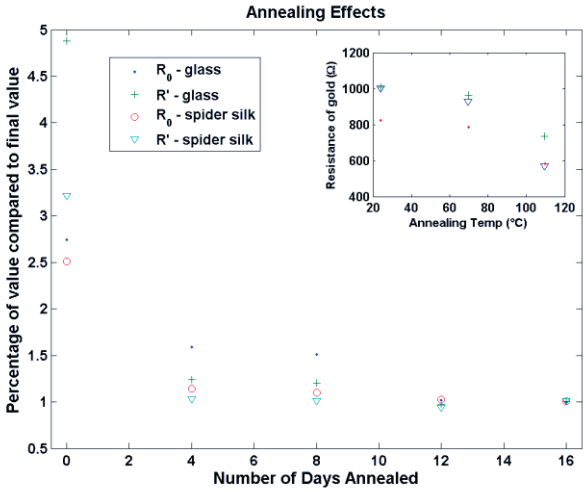


Figure 5. Annealing effect on property of interest as a function of time and resistance of gold on glass fiber change due to annealing temperature (inset).

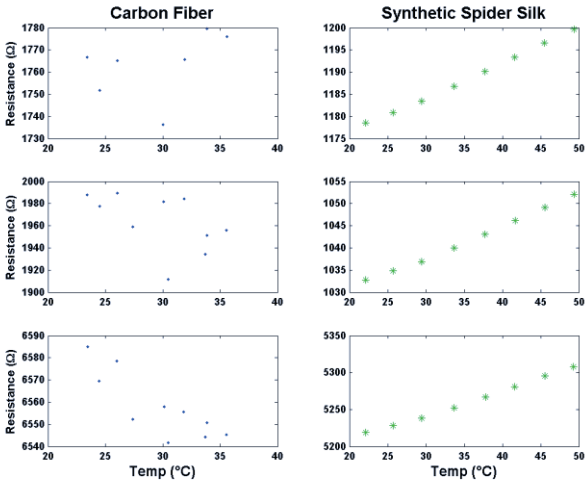


Figure 6. Resistance-temperature calibrations showing poor calibrations with carbon fiber (left) and good calibrations with synthetic spider silk (right).

Following the path on the flow chart of the short rise time, the 3ω method takes advantage of ac heating and improving signal-to-noise ratio by locking-in to the 3rd harmonic of the heating source. Two options are available to provide the modulated heating required for the 3ω method, a current source and a voltage source. The selection of which source to use is determined mostly by what equipment is available, but both want the amplitude of the measured 3ω voltage to be between 1 and 5 μV for good signal while keeping temperature rises small. Regardless of what is used, it must be able to be sinusoidally modulated and have the ability to plug in to a lock-in amplifier to provide a reference signal to lock-in to the 3rd harmonic. If both are available, then the voltage source is suggested because

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it can give slightly better results, although it requires the use of a bridge and a conversion (Eq. 11) to relate the measured bridge voltage to the true 3ω voltage of the fiber. The bridge can then be balanced or not balanced to cancel out the 1ω signal, with the option of canceling out the 1ω signal providing greater accuracy. However, to calculate the thermal conductivity from the amplitude of this 1ω-canceled signal, a regression based on the ratio of sample resistance to total bridge resistance – $R_0/(R_0+R_b)$ – is suggested [1] to extrapolate the thermal conductivity of each fiber if there was no resistance (an ideal current source). For larger sample resistances, this is less of an issue. Finally, both sources can then use Eqs. 8 (curve-fit), 9, and 10 for thermal conductivity, thermal diffusivity, and specific heat, respectively. The applicability of these models are based on the frequency that the sources is modulated at, using both the amplitude and phase of the signal, with the intent that the phase must be between 0° and -90°.

TET Considerations

Following the path on the flow chart of the long rise time, the TET method is, essentially, a 4-wire resistance measurement at high enough currents that the temperature and resistance of the wire changes. A constant current is applied to the fiber and the voltage response is recorded by a high speed DMM. Because of the constant current, the resistance response is directly related to the voltage response, and by means of a good calibration, the average temperature response is directly related to the resistance response. The range of currents used for the probing are selected so that at the low end, the transient curve is barely observed while at the high end, a good signal to noise ratio is obtained while still not letting the temperature rise of the fiber get too high (5°C max). The typical ranges are between 10μA-10mA, depending on the resistance of the sample. Finesse is needed to find the transient curve at times because the absolute change in voltage is very small compared to the magnitude of the voltage measured. This is seen in [4] (Figures 4 and 5) and leads to an important consideration of what the R_0 of the fiber is for use in the conductivity models (Eqs. 2, 4, and 6). Three routes are considered. The first is by using a very small constant current so that no heating occurs and measuring the resistance of the sample by the 4-wire method, separate from the TET test. The issue is that this R_0 resistance, when used in the model, is too low and produces errors in the conductivity calculation, hence this is an invalid way. The second is a simple way, where the voltage is noted from the initiation of Joule heating and this voltage is then used to determine the R_0. For noisy data sets, this is further improved during the normalization of the voltage rise by slightly varying the starting points of the data so that upon averaging of the data, the expression $(\Delta V-V_0)/(V_s-V_0)$ yields a value near zero at the first point on the transient curve, where t=0. The third is a more complex way that gives similar results to the simple Joule heating initiation route. The reduced model for k is used (Eq. 4) with the R_0 determined by the low-current DMM, and k is solved for the same fiber but at the different currents, which results in different k values. A regression process can then be done to vary the R_0 (which is the same at each current) until the rms error between the different k values is minimized.

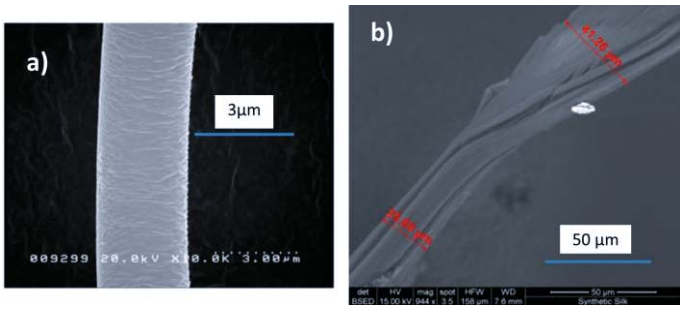


Figure 7. SEM images of natural (a) and synthetic spider silk (b) showing the variation in sample geometry uniformity.

Once the data has been acquired and the correct R_0 has been selected, the uniformity of the sample geometry then determines which properties can be determined. Uniform samples (Fig. 7a) can get both k and α, while non-uniform (Fig. 7b) can only accurately get thermal diffusivity. However, the approach is very similar for both geometries. If the emissivity of the sample is accurately known, then Eq. 6 can be used. However, this is rarely the case because of emissivity's dependence on many factors; therefore additional routes must be followed. The simplest (and the route that must be done if there is a poor calibration) is to assume that the H_e^2 of the fibers on the same mount is the same. With this assumption, the only difference between the samples is the square of their length. The reduced models (Eqs. 3 and 4) can then be applied to get the property of each fiber, and then these values can be plotted as a function of L^2. As shown in Fig. 8, there exists a linear relationship because as the fibers get longer, the amount of radiation heat loss increases and the thermal property over-estimation increases. This relationship can then be exploited and an extrapolated value of a "zero-length" fiber (where radiation heat losses do not occur) can be determined as the intercept of the linear regression. This value matches well with literature values for platinum and gold-coated glass when applied and also matches well with the values measured by the full models. An additional check on this can be done when a regression procedure is done to simultaneously fit for both the thermal property and the H_e^2 of the sample. With this effective heat transfer coefficient, a k value can then be calculated by rearranging of the expression for H_e^2, neglecting non-constant heating, and this is a potential way to overcome the issues when a bad calibration is observed. However, this method has not been verified.

In the case where the calibration is good and sample geometry is uniform, the full models can be applied (Eqs. 2 and 3) and give the additional benefit of improved accuracy by inclusion of radiation effects (non-avoidable), convection heat loss (avoidable through vacuum), and non-constant heating

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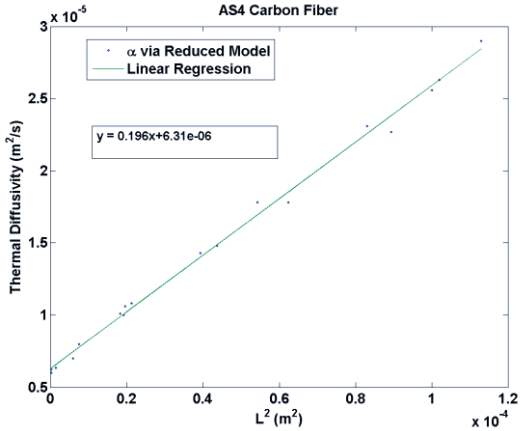


Figure 8. Thermal diffusivity results by reduced TET model for Hexcel AS4 carbon fiber, showing linear dependence on length and the regression result that is used for the improved "zero-length" property measurement.

(avoidable by reducing the temperature rise experienced by the fiber). Because of the dependency of k in both equations, implicitly in the k expression and explicitly in the α expression (because it is incorporated in the H_e^2 term), it is suggested to use the k value determined by the reduced model as an initial guess for Eq 2 and then use that improved value in Eq 3. These full model values can also be plotted as a function of the square of the length and have the same "zero-length" extrapolation performed, but with little additional accuracy added. The benefit of these different methods (full model, reduced model with radiation correction – modified, and reduced model with regression to "zero-length" fiber) is that they give very similar results, as evidenced by Fig. 9, where the reduced model k was used to get an improved full model k, which was used for the full model α (blue dots), then the modified α was determined based on a regressed error term from the reduced model (green circle), and the "zero-length" value determined by the linear regression of the reduced model is also given (red dashed line).

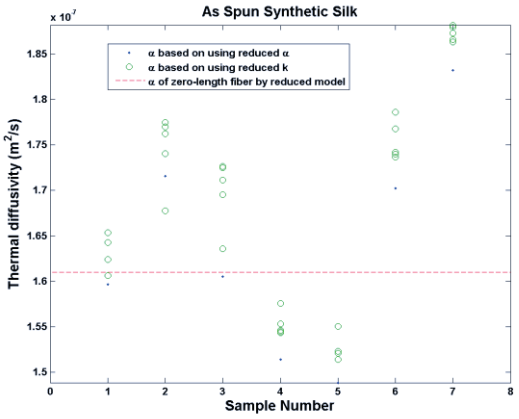


Figure 9. Thermal diffusivity of synthetic spider silks for the same sample mount using different models.

FUTURE WORK

Because of the long lead time for a sample measurement (several weeks from fiber reception to final property measurement) and the likelihood of several of the fibers on a single mount being broken during the entire process, there exists a need for a thermal property measurement that has a faster turn-out rate. One proposed solution is a non-contact method that uses the spectral features of fluorescent dyes coated on the surface of a fiber as a temperature sensor via training of a neural network based on the spectra at known temperatures. The decay of a photothermally induced thermal wave can then be measured and related to the thermal diffusivity of the fiber. The lead time for this method is short because the fibers can easily be coated with the fluorescent dye, then placed in the focal plane of the laser path, the spectral features can be measured at calibrated temperatures, and then the scanning of the fiber for the thermal wave decay can all happen in the space of several days, rather than weeks.

CONCLUSIONS

Previous thermal property results of materials used for validation and spider silks based on the TET and 3ω methods have been compiled and compared to new results from synthetic spider silks with improved mechanical (and thermal) properties. The processed synthetic silks have improved properties compared to the unprocessed silks, but are not at the same magnitude as the natural silks from which they are derived. A detailed description of the processes used to obtain an accurate thermal property measurement are presented in a flow chart and explanatory text to aid in the measurement of the thermal properties of thin fibers. Future work of a non-contact method for thermal diffusivity measurements based on fluorescence-based temperature sensing is briefly presented.

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NOMENCLATURE

α	Thermal diffusivity
α_T	Temperature coefficient of resistance
γ	Non-constant heating influence term
ε	Emissivity
ρ	Density
ρ_e	Electrical resistivity
σ	Stefan-Boltzmann constant
ξ	Radiation influence term
ω	Modulation frequency
3ω	Third harmonic signal
C_p	Volumetric specific heat
D	Fiber diameter
DC	Mean value of waveform
DMM	Digital multimeter
e_{kb}	Radiation correction factor
h_c	Convection heat transfer coefficient

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H_e	Effective heat transfer coefficient
h_r	Radiation heat transfer coefficient
h_c	Convection heat transfer coefficient
i	Imaginary number
I	Probing current
I_0	DC current amplitude
IR	infrared
k	Thermal conductivity
L	Length of fiber
m	Summation index, or complex factor
PCB	Printed circuit board
R_b	Total bridge resistance
R_s	Steady state resistance
R’	Resistance-temperature slope
R_0	Initial fiber resistance
t	Time
TET	Transient Electrothermal Technique
ΔT	Transient average temperature rise
ΔT_s	Steady state average temperature rise
T_0	Initial fiber temperature
V_b	Measured bridge voltage
V_3	Measured 3ω voltage
ΔV	Transient average voltage rise
ΔV_0	Initial average voltage rise
ΔV_s	Steady state average voltage rise

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